Neural Stiff-Chemistry Surrogates in DeepFlame: Implementation Pitfalls, Coupled-Solver Safeguards, and Target Semantics for H_2 and C_2H_4

Project draft generated from experiment record

April 26, 2026

Abstract

Neural surrogates for stiff chemistry are increasingly attractive for reacting-flow simulation, but their practical value depends on more than offline fit. This manuscript studies an EFNO-oriented implementation and deployment workflow for H_2 and C_2H_4 chemistry inside DeepFlame, with emphasis on coupled usefulness under explicit runtime contracts. Four conclusions are supported by the present evidence. First, implementation details matter enough to change scientific interpretation: on the H_2 benchmark, a transformed-state decode bug made self-rollout appear much less stable than it became after correction. Second, offline success does not transfer automatically to CFD deployment: DeepFlame-coupled H_2 runs fail first through thermodynamic reconstruction, which makes operating windows, fallback logic, and guard policies central parts of the method. Third, a credible partial deployment story is now available for H_2: the corrected branch, combined with frozen-temperature gating and an optional logistic risk guard, yields reproducible retention-oriented and safety-oriented operating modes. Fourth, the dominant remaining C_2H_4 limitation is not simply model capacity but target semantics: the best current mixed-data model survives to the stock runtime horizon, yet still over-drives cooler active cells, exhibits hot-regime source-term sign errors, and suppresses key intermediates, while naive chemistry-proxy relabeling fixes semantics too bluntly to preserve the solver-relevant manifold. The paper is therefore framed as a partial-results implementation study whose main contributions are diagnostic: contract-sensitive reproduction, deployment-facing validation, and explicit evidence that target semantics govern the remaining C_2H_4 gap.

1 Introduction

Conventional stiff chemistry integration remains one of the main computational bottlenecks in high-fidelity reacting-flow simulation. Recent deep-learning surrogates offer a plausible alternative, but the question that matters for CFD practice is not only whether a network can fit one-step chemistry data offline; it is whether the resulting model can survive coupled deployment under the thermodynamic, postprocessing, and runtime constraints of a real solver [2, 1, 4]. That distinction motivates the present study.

This paper examines an EFNO-oriented workflow for H_2 and C_2H_4 chemistry in DeepFlame, but its main contribution is not a blanket claim that one neural architecture has won. Instead, the work asks a more operational question: what has to be true for a learned stiff-chemistry surrogate to become solver-usable in CFD? The evidence developed here suggests three necessary ingredients. First, the implementation contract must be correct, especially when states are transformed before learning. Second, evaluation must be deployment-facing, because offline rollout scores can still disagree with solver behavior once postprocessing and thermodynamic reconstruction are applied.

Third, the target labels must reflect the chemistry one actually wants the network to learn; otherwise even runtime-stable models can converge to the wrong reactive manifold.

Those themes appear differently in the two fuels studied here. For H_2, the strongest current result is a deployment-oriented one: once a key transformed-state decode bug is fixed, the corrected branch becomes a credible basis for guarded DeepFlame deployment, and the decisive design choices become fallback logic, activation threshold, and runtime contract rather than one more offline sweep. For C_2H_4, by contrast, the strongest current result is still diagnostic: after the runtime bridge is stabilized, the remaining error is tightly structured around target semantics, intermediate suppression, and source-term sign errors rather than mere survivability. The paper is therefore written as a rigorous partial-results study whose value lies in identifying reproducible operating regimes, failure modes, and design rules rather than in claiming solved surrogate chemistry replacement across all coupled settings.

2 Background and related work

The manuscript is grounded in the EFNO paper by Weng et al. [3] and a local-paper crosswalk covering data generation, augmentation, and coupled-validation issues from related combustion-chemistry learning papers [2, 5, 1, 4]. Across these papers, a recurring theme is that model architecture alone is not decisive: data semantics, postprocessing contracts, augmentation strategy, and a posteriori validation all materially affect conclusions.

3 Reproducible workflow and software stack

The implementation workflow combines PDF extraction and structured notes, DFODE-kit extensions for EFNO-style training and evaluation, and DeepFlame export bridges for MLP and FNO-like surrogates. The software path is deliberately aligned with the emerging DFODE-kit/DeepFlame ecosystem [1, 2]: data are prepared from canonical or case-aligned thermochemical states, models are trained and evaluated offline with explicit rollout and physical-consistency metrics, and promising checkpoints are exported into DeepFlame for coupled testing. All substantive experiments are documented in the project docs site together with scripts and artifact paths.

4 Methods and evaluation protocol

The study is organized as a staged implementation-and-deployment workflow rather than a single train/test exercise. Table 1 summarizes the main benchmark and case families used in the paper so that the offline and coupled evidence can be read against the correct mechanism, split, and runtime contract.

The offline H_2 benchmark uses autoignition trajectories generated from a seven-species hydrogen mechanism, with holdout splits defined by initial-condition trajectories rather than random sample shuffling so that rollout claims reflect unseen trajectory behavior rather than interpolation within the same paths. Models are compared with fixed seeds and evaluated on one-step thermochemical error, long-horizon rollout error, and physical-consistency metrics including element-mass and mass-sum deviations. This choice follows directly from the EFNO paper's emphasis on autoregressive prediction under unseen conditions [3], but the present implementation also makes the evaluation contract explicit because several apparently strong branches change rank once deployment-facing postprocessing is enforced.

Table 1: Compact overview of the main benchmark/case families used in the manuscript.

Study block	Data / mechanism	Split or deployment contract	Primary reported metrics
Offline H_2 holdout	7-species ES80 autoignition trajectories	trajectory holdout; autoregressive rollout	1-step/rollout MAE; element and mass consistency
Coupled H_2 Burke	9-species Burke DeepFlame case	exported checkpoints under preserve-last runtime contract	HP-failure risk; learned/fallback fractions; operating windows
Stock C_2H_4 baseline	DeepFlame np=8 GPU stock example	trusted baseline to 5×10^{-6}	runtime survival; matched-mesh T/p/$\dot{Q}$ sanity
Best learned C_2H_4	*dp*100 case-pairs + canonical@0.2 mix	batched FNO bridge against stock horizon	matched-mesh $\dot{Q}$/T/p/species comparison at 5×10^{-6}

For DeepFlame-coupled H_2, the target case is the Burke nine-species configuration rather than the earlier seven-species offline mechanism. That mechanism/species-count alignment became a hard requirement once export began. Coupled evaluations are therefore performed on Burke-aligned exported checkpoints and interpreted under the same species postprocessing contract used by the runtime bridge. In practice, this means distinguishing the project-side closure reconstruction from the DeepFlame-facing "preserve-last" rule, where the last species is preserved from the current state and the predicted main species are renormalized. This contract choice materially compresses some offline ranking gaps and is therefore part of the method, not an implementation footnote.

For C_2H_4, the baseline is a stock-style DeepFlame run in the trustworthy GPU regime (‘mpirun -np 8‘, ‘coresPerNode 8‘) carried to 5×10^{-6}. Learned-model experiments are only interpreted after this runtime baseline and after the exported FNO bridge was repaired to batch active-set inference, because earlier reduced-rank GPU crashes and late out-of-memory failures showed that runtime pathologies could otherwise be mistaken for model failures. The main C_2H_4 learned datasets then span three increasingly deployment-relevant families: crude case-pair labels extracted from CFD states, filtered/mixed case-pair subsets such as *dp*100, and canonical/augmented data motivated by nearby literature on reactive-region coverage [2, 5, 1]. During this stage we also confirmed that DFODE-kit's documented DeepFlame-backed one-dimensional freely propagating flame workflow can be instantiated for the C_2H_4 Wu24sp mechanism and sampled into HDF5, which creates a more faithful future starting point for Xiao-style one-dimensional flame sampling than the earlier Cantera-only canonical prototype. A direct manifold comparison then showed that the raw DeepFlame one-dimensional sample is a better physical backbone than the Cantera-only canonical sample in pressure support, but it still under-covers the reactive intermediate manifold that appears in the stock active set; conversely, the current canonical augmentation over-corrects in the opposite direction. Applying Xiao-style interpolation and constrained perturbation directly on top of the DeepFlame-sampled one-dimensional manifold moved the current-state support into a more balanced regime and produced a first deployed model that survived to 10^{-6}, but the learned chemistry still deleted key intermediates such as C_2H_5, C_2H_3, CH_2CHO, and CH_2CO. A follow-on mixed-label test combining *dp*100 with a small one-dimensional augmented add-on likewise preserved early runtime survival without rescuing this chemistry gap. Importantly, a matched DeepFlame C_2H_4 reference run with standard chemistry integration (‘torch off‘) showed that the stock learned example remains much closer to the in-loop chemistry baseline at 10^{-6} than these current project surrogates: the stock model preserves the key intermediate manifold within order-one factors, whereas the current one-dimensional and mixed-label project models suppress those intermediates by many orders of magnitude while over-driving $\dot{Q}$. Replacing the current Box–Cox state-delta species target

with a Xiao-style power transform on physical ΔY materially improved bulk deployment behavior for the mixed $dp100$+one-dimensional case: the mean active-region $\dot{Q}$ ratio versus the CVODE baseline collapsed from about $8.6\times$ to 8.5×10^{-4}, while the mean temperature and pressure errors also dropped. However, this first power-delta deployment still left the key intermediate manifold effectively extinguished. A first lightweight attention block added on top of the spectral token path then showed that architecture also matters, but in a selective way: some collapsed channels such as C_2H_3 and CH_2CHO moved upward by several orders of magnitude, while others such as C_2H_5 and CH_2CO worsened, and the bulk $\dot{Q}$ behavior became less damped again. Adding a first species-aware weighting profile on top of that attention model partially rebalanced the tradeoff once more: the weighted model pulled $\dot{Q}$ back toward the safer damped regime and greatly improved CH_2CHO relative to the earlier power-delta baseline, but it still did not recover the full intermediate manifold and gave back most of the C_2H_3 gain from unweighted attention. This strengthens the case that one-dimensional DeepFlame augmentation is useful as a solver-native manifold source and that both target transforms and token-interaction structure matter substantially, but not yet as a drop-in labeled-data fix without further regime-aware semantic refinement.

Across both fuels, the manuscript prioritizes deployment-facing metrics over raw fit alone. Offline results are summarized by one-step MAE and rollout MAE, but coupled H_2 decisions are further driven by HP-reconstruction failure fractions, learned-vs-fallback active-cell fractions, and threshold/guard operating windows. Coupled C_2H_4 decisions are driven by survival to the stock baseline horizon together with matched-mesh comparisons of $\dot{Q}$, temperature, pressure, and key species. This method deliberately treats negative results—for example, preserve-last rebaselining, pure chemistry-proxy early failure, and random partial relabeling regression—as evidence about evaluation contracts and target semantics rather than as dead ends.

5 Offline H_2 reproduction and implementation diagnosis

The offline H_2 study began as a reproduction exercise, but the main result became methodological rather than purely comparative. Seeded baseline runs initially suggested that rollout-aware EFNO-style training could outperform the supervised MLP reference by a wide margin on long-horizon metrics. However, branch-by-branch diagnosis of the second-step feature path showed that the apparent instability of self-fed rollout was dominated by a transformed-state species decode bug: species updates were being combined across incompatible state spaces. After correcting the update rule to act in the proper transformed space, the empirical picture changed qualitatively. Self-rollout branches that had previously looked intrinsically unstable became the strongest rollout performers, while teacher-forced rollout remained competitive rather than dominant.

Figure 1 makes the scale of this change explicit. The correction was not a small numerical cleanup; it is the difference between concluding that self-feeding is fundamentally harmful on this benchmark and concluding that a large part of the previous failure was implementation-induced. This is an important paper-level point because it reframes "reproduction" as contract-sensitive implementation diagnosis: the benchmark is useful not only for ranking models, but for exposing which conclusions are robust to the details of transformed-state decoding and autoregressive feature construction.

6 DeepFlame-coupled H_2: from failure to guarded deployment

The coupled H_2 study then tested whether the improved offline story survived contact with a real CFD runtime. It did not, at least not directly. Exported Burke-aligned checkpoints loaded

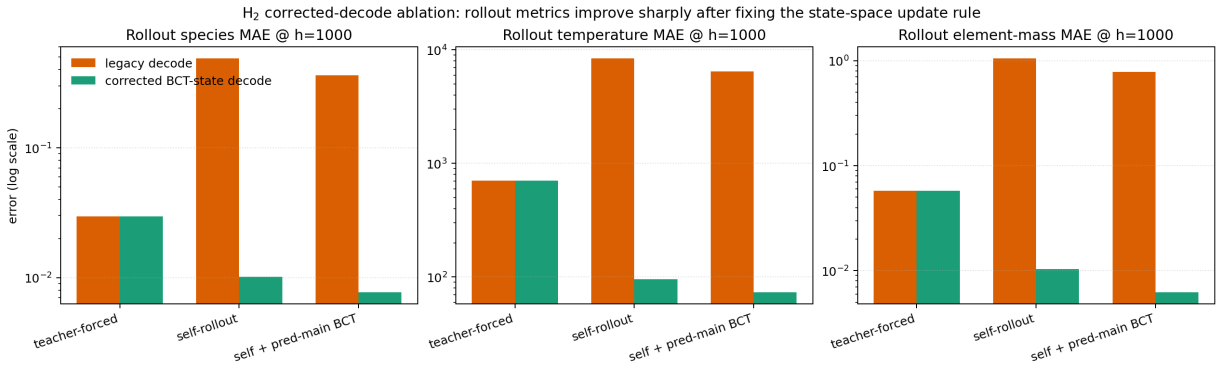

Figure 1: H_2 legacy-vs-corrected decode ablation on rollout metrics at horizon 1000. Fixing the transformed-state update rule changes the empirical conclusion most strongly for the self-rollout branches, supporting the claim that a large part of the earlier instability was implementation-induced rather than intrinsic to self-feeding.

Table 2: Selected H_2 coupled-deployment operating points for the corrected branch in DeepFlame.

Mode	Learned frac. at 2×10^{-5}	Cum. fallback frac. at 2×10^{-5}	First time > 95% fallback
FT550	0.000	0.516	1.4e-05
FT600	0.081	0.373	–
FT650	0.294	0.361	–
FT650+risk@0.5	0.241	0.381	–

correctly and began evolving the case, but both supervised and rollout-aware branches encountered thermodynamic failure through HP reconstruction after only a few microseconds. That failure mode mattered because it shifted the central question away from offline accuracy and toward solver compatibility: the immediate problem was no longer whether the network predicted species well in isolation, but whether the coupled update remained thermodynamically reconstructable.

This failure, however, became informative rather than terminal. Snapshot analysis, written-field inspection, and explicit HP-risk reconstruction turned the crash into a measurable operating-window problem. Hybrid fallback policies, frozen-temperature gating, and a logistic risk guard were then introduced as deployment mechanisms rather than ad hoc hacks. Figure 2 shows how threshold and guard choices reshape the coupled operating regime over time. Table 2 summarizes representative operating points for the corrected branch. In the present evidence base, 650 K gives the strongest long-horizon single-threshold retention, while the logistic risk guard at threshold 0.5 provides a more safety-oriented mode with lower HP-failure exposure at some cost in learned participation.

The H_2 evidence therefore supports a reviewable claim stronger than "the corrected branch works better." It supports the narrower but more consequential claim that deployment-facing chemistry surrogates require contract alignment and explicit thermodynamic safeguards, not just better offline loss values. In other words, coupled usability emerges from the interaction between model, postprocessing rule, activation threshold, and fallback logic; no single offline metric captures that interaction by itself.

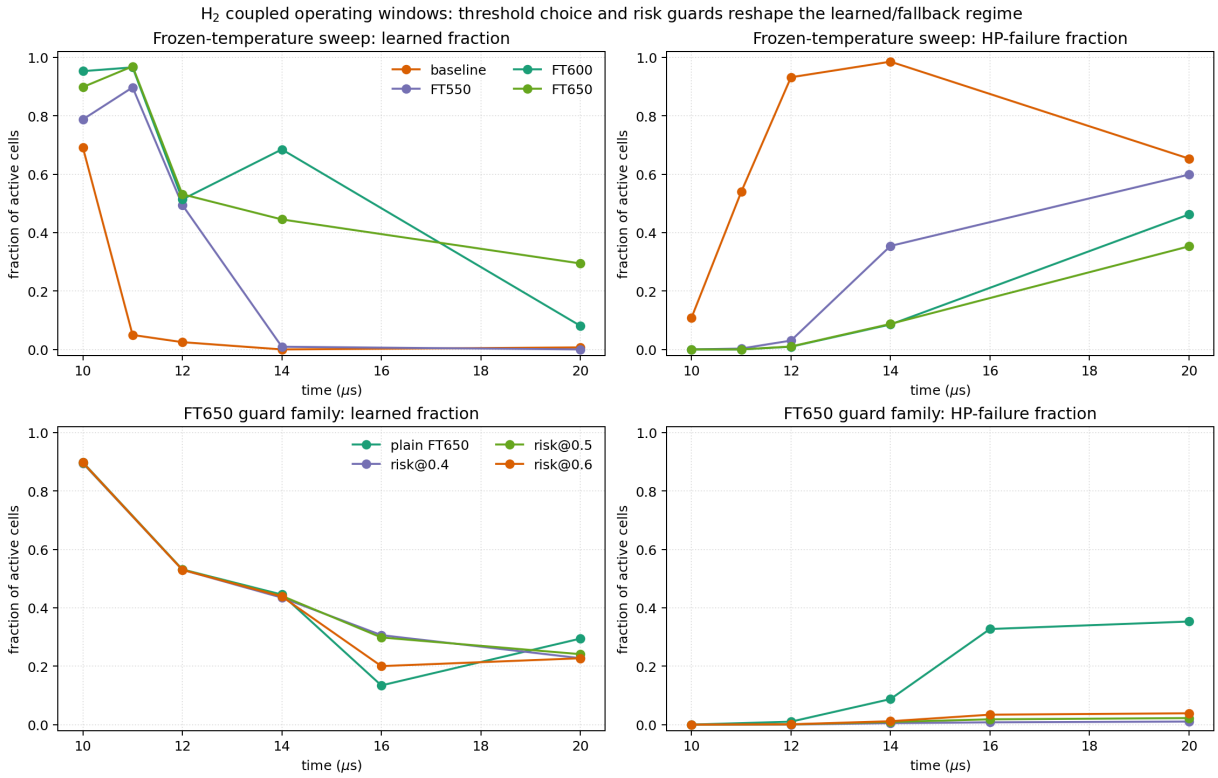

Figure 2: H_2 coupled operating windows from the corrected branch. Top row: frozen-temperature sweep showing learned-cell retention and HP-failure fractions over time. Bottom row: FT650 risk-guard family. The figure makes two manuscript-level points explicit: raising the activation threshold delays collapse materially, and the logistic risk guard trades some learned participation for much lower HP-failure incidence.

7 C_2H_4 baseline establishment and runtime bridge diagnosis

For C_2H_4, the first task was not model improvement but baseline legitimacy. Early reduced-rank CPU/GPU experiments were useful for diagnosing packaging and communication failures, yet they were not trustworthy scientific baselines because the stock example was clearly written around a different runtime regime. The program therefore re-established the case in the stock-style ‘np=8‘ GPU configuration and only then began treating the resulting trajectory as the deployment target. This distinction matters for the paper narrative: before a trustworthy stock baseline exists, it is too easy to misattribute infrastructure failure to chemistry-model failure.

The same logic applied to the learned runtime bridge. The first case-aligned FNO models could survive farther than the earliest homogeneous-data smoke baseline, but late out-of-memory failures and zero-$\dot{Q}$ written fields showed that the deployment path itself was still contaminating interpretation. Batching active-set inference repaired that bridge. Only after those runtime issues were removed could the remaining C_2H_4 discrepancies be interpreted as evidence about target quality rather than infrastructure. This emphasis on a posteriori validation follows the practice advocated in recent combustion-surrogate studies, which consistently show that apparently strong offline models can still fail once coupled back into CFD [2, 5, 4].

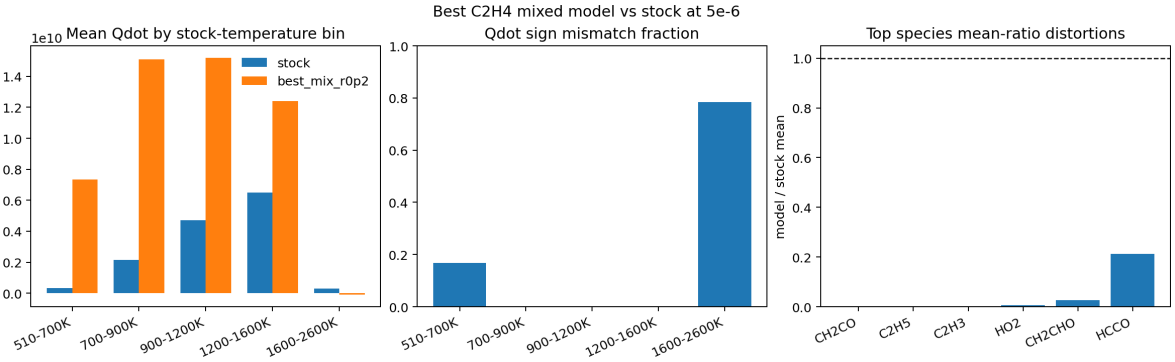

Figure 3: Matched-mesh comparison between the current best mixed C_2H_4 model and the stock DeepFlame baseline at 5×10^{-6}. The figure highlights temperature-binned source-term mismatch and the strongest mean-ratio species distortions.

Table 3: C_2H_4 deployment-facing summary at 5×10^{-6} for the stock baseline, the current best mixed model, and a naive partial chemistry-proxy relabeling ablation.

Model	Mean $\dot{Q}$	$\dot{Q}$/stock	$p_{\max}$ (Pa)	$T_{\min}$ (K)	mean $\lvert\Delta T\rvert$ (K)
Stock DeepFlame	1.62e+07	1.000	1.02e+05	499.249	0.828
Best mixed ($dp100 + canon@0.2$)	2.51e+07	1.551	1.03e+05	499.159	0.927
Partial chem-proxy (10%)	-9.88e+08	-60.952	1.03e+05	492.524	9.153

8 C_2H_4 data semantics: what the current best model still gets wrong

Once the C_2H_4 runtime path was stabilized, the central question shifted from "can a learned model survive?" to "what chemistry is the current label path actually teaching?" The strongest current data result is a calibrated mixed dataset that combines the solver-aligned $dp100$ case-pair backbone with a moderate canonical augmentation component. This mixed model survives to 5×10^{-6} and is substantially better than earlier baselines, which is already an important positive result for the paper. But it does not fail in a random or weakly structured way. The best-model-vs-stock anatomy shows a coherent residual pathology: cooler active cells are over-driven, the hot bulk exhibits widespread source-term sign errors, and several late intermediates are still heavily suppressed. The mixed-data direction itself is motivated by nearby literature showing the value of canonical flame manifolds, interpolation-based densification, and physics-aware perturbation for building thermochemical training sets that cover reactive regions more effectively [2, 5, 1].

The next diagnostic step showed that the target-semantics hypothesis is real in the data itself: one-step chemistry-proxy relabeling of case-pair current states produces materially different targets, especially on key intermediates. Table 4 summarizes the most distorted active-region species in the current best mixed model, while Table 5 shows that the original CFD next-state labels differ materially from chemistry-proxy relabels on a sampled subset. However, deployment tests also showed that semantics-only fixes are not enough in isolation. A tiny pure chemistry-proxy model fails catastrophically by 4×10^{-7}, while naive partial replacement of 10% of labels preserves runtime survival but badly degrades solver-facing quality.

Table 4: Largest active-region species distortions for the current best mixed C_2H_4 model relative to stock at 5×10^{-6}.

Species	Stock mean	Model mean	Model/stock
CH2CO	2.12e-04	2.95e-17	1.39e-13
C2H5	1.50e-05	2.22e-16	1.47e-11
C2H3	5.81e-05	3.45e-10	5.95e-06
HO2	5.46e-05	3.64e-07	0.007
CH2CHO	6.63e-06	1.77e-07	0.027
HCCO	6.01e-06	1.28e-06	0.212

Table 5: Largest mean-ratio shifts between original CFD next-state labels and chemistry-proxy relabels on a 5k-row C_2H_4 subset.

Species	Original next mean	Chemistry-proxy next mean	Chem/original
HCCO	3.82e-07	1.34e-05	35.146
C2H5	5.13e-06	2.12e-07	0.041
CH2CHO	2.54e-06	1.08e-05	4.269
CH2OH	1.38e-06	5.13e-06	3.713
C2H3	3.90e-05	1.03e-04	2.651
CH2CO	6.24e-05	1.64e-04	2.621

Table 3 makes the current C_2H_4 claim boundary explicit. The best mixed model is reviewably better than prior baselines, but naive partial chemistry-proxy relabeling is not yet a deployment improvement. Tables 4 and 5 sharpen the interpretation in an important way. Together they suggest that the remaining gap is not primarily one of network capacity or another nearby mixing ratio; it is a semantic gap between the full-CFD labels available cheaply and the chemistry-only evolution that would better preserve the intermediate manifold. The negative controls are therefore informative: pure chemistry-proxy relabeling fixes semantics too aggressively and loses the solver-relevant manifold, while random partial relabeling preserves the manifold but injects the new semantics in the wrong places.

A new targeted relabeling scout now makes that selective-semantics direction more concrete. On a late-window $dp100$ subset, a regime-targeted selector that concentrates relabel budget on cooler/intermediate-rich rows raises the selected intermediate-species mass from 5.36×10^{-4} for a random 3k subset to 3.13×10^{-3}, while shifting the mean temperature from 2213 K to 1399 K. One-step chemistry-proxy relabeling on this targeted subset then shows an even larger semantic gap than the earlier random early-window scout, with mean absolute T_{next} difference about 656 K and order-of-magnitude shifts on species such as HCCO and C_2H_5. Deployment tests then show that row choice and replacement strength interact strongly. A targeted 10% partial-relabel model collapses by 4×10^{-7} with severe thermodynamic excursions, while lowering the same targeted strategy to 2% delays failure to 9×10^{-7} but still does not recover solver-usable behavior. This refines the forward claim further. Selection quality matters, but simply injecting stronger chemistry-proxy semantics through direct substitution is not enough; replacement strength and staging likely matter as much as row choice. This supports the more specific forward hypothesis that semantics must be improved selectively rather than globally.

9 Discussion

The present evidence supports several broader lessons that cut across both fuels. First, transformed-state implementation details can dominate empirical conclusions, which reinforces the need to treat reproduction as implementation diagnosis rather than a narrow hyperparameter chase [3]. The H_2 decode-contract result is the clearest example: a bug in how states were updated across transformed spaces could easily have been misread as a limitation of rollout-aware learning itself.

Second, offline and deployment-facing evaluations can disagree substantially because of post-processing and thermodynamic contract issues, in line with the broader combustion-surrogate literature's emphasis on a posteriori validation [2, 4]. In this study, the preserve-last versus closure distinction changes offline H_2 rankings, and HP-reconstruction risk governs whether an apparently good checkpoint is solver-usable at all.

Third, the two fuels expose different limiting mechanisms. For H_2, the strongest current challenge is deployment control: the offline branch can be made good enough that guard design, threshold choice, and runtime contract become decisive. For C_2H_4, by contrast, the strongest current challenge is still target construction. Once runtime pathologies were removed, the residual errors remained tightly structured around source-term sign, intermediate suppression, and chemistry-versus-CFD label mismatch. That pattern argues against spending the next effort on another small architecture or ratio sweep.

Taken together, these lessons define the manuscript's central claim boundary. The paper does not show that learned chemistry has already replaced conventional integration in all relevant CFD settings. It does show that careful implementation, deployment-aware validation, and explicit treatment of target semantics can move the question from generic promise to concrete operating regimes, failure modes, and reproducible design rules.

10 Conclusion

The current project state is strong enough for a reviewable partial-results paper because it supports a concrete, bounded claim. For H_2, corrected implementation plus deployment safeguards yields a credible coupled story: once the transformed-state update contract is fixed, learned chemistry can be kept solver-usable in DeepFlame through explicit operating-window design rather than through offline accuracy alone. For C_2H_4, the strongest result is equally concrete but more limited: a calibrated mixed-data model now reaches the stock runtime horizon, yet its residual errors remain organized around target semantics, especially intermediate suppression and source-term sign mismatch, which shows that further progress depends more on selective relabeling and target design than on another small architecture sweep.

The broader conclusion is methodological. Learned stiff-chemistry surrogates should not be judged only as regressors on isolated datasets. They should be treated as coupled numerical components whose usefulness depends on implementation correctness, postprocessing contracts, thermodynamic reconstructability, and the meaning of the target labels themselves. Under that standard, the present study does not claim that neural surrogates have already replaced conventional chemistry integration across all relevant CFD settings. It does claim that a deployment-aware workflow can turn that broad ambition into reproducible operating regimes, interpretable failures, and actionable design rules for the next generation of H_2 and C_2H_4 surrogate-chemistry models.

Immediate remaining tasks before submission

1. Continue tightening the prose into a venue-ready narrative, especially the introduction and conclusion.
2. Decide whether to add a final manuscript-grade C_2H_4 failure-boundary figure or keep the current claim boundary text-only.
3. Replace naive semantics-only C_2H_4 relabeling with a more selective semantics-improvement experiment, or scope that step explicitly as future work.
4. Add any venue-specific formatting changes once a submission target is chosen.

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